CO-1 AT-2 OUTPUT PREDICTION

Name : Ranjitha
Reg No : 192421285

Course Code : CSA 0311
Course Name : Data Structure
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Question 1 - Second Largest element

Your Code

```c
#include<stdio.h>
int main(){
    int n,i;
    printf("Enter the number of element: ");
    scanf("%d",&n);
    int a[n];
    printf("Enter the elements: \n");
    for(i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    int largest = a[0],second = a[0];
    for(i=1;i<n;i++){
        if(a[i]>largest){
            second =  largest;
            largest = a[i];
        }else if(a[i]>second && a[i]!= largest){
            second =a[i];
        }
    }
    printf("Second largest element = %d", second);
    return 0;
}
```

Input

```
10
25
8
40
30
```

Output

```
Enter the number of element: Enter
the elements:
Second largest element = 30
```

Question 2 - largest and smallest elements

Your Code

```c
#include<stdio.h>
int main(){
    int n,i;
    printf("Enter the number of element: ");
    scanf("%d",&n);
    int a[n];
    printf("Enter the elements: ");
    for(i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    int largest = a[0];
    int smallest = a[0];
    for(i=1;i<n;i++){
        if(a[i]>largest)
            largest = a[i];
        if(a[i]<smallest)
            smallest = a[i];
    }
    printf("Largest element = %d\n",largest);
    printf("Smallest element = %d\n",smallest);
    return 0;
}
```

Input

```
10
15
20
25
30
```

Output

```
Enter the number of element: Enter
the elements: Largest element = 30
Smallest element = 10
```

Question 3 - first and last occurrence elements

```c
#include<stdio.h>
int main( ){
    int n, i, key;
    printf("Enter the number of element: ");
    scanf("%d",&n);
    int a[n];
    printf("Enter the elements: \n");
    for(i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    printf("Enter the element to search: ");
    scanf("%d",&key);
    int first = -1, last =-1;
    for(i=0;i<n;i++){
        if(a[i] == key){
            if(first == -1)
                first =i;
            last= i;
    }
    }
    if(first==-1)
        printf("Element not found");
    else{
        printf("First occurrence = %d\n", first);
        printf("Last Occurrence = %d\n",last);
    }
        return 0;
}
```

```
40
50
70
80
```

Output

```
Enter the number of element: Enter
the elements:
Enter the element to search:
Element not found
```